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An Ising-like Model for Monolayer-monolayer Coupling in Lipid Bilayers

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We have proposed the Ising bilayer model to study the domain growth dynamics in lipid bilayers. Interactions within and between layers are adopted from recent experimental and theoretical data. We investigate the effects of the mismatch area on the domain coarsening dynamics in both symmetric and asymmetric lipid bilayers. To explore domain coarsening, we used the Monte Carlo (MC) method with a standard Kawasaki dynamics to simulate the systems. The results show that domains on both layers grow following a power-law and that the domains grow slower when the mismatch areas are increased.

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I. INTRODUCTION

Synthetic lipid membranes, such as giant unilamella vesicles (GUVs), exhibit phase separation at temperature below their critical temperatures. The lipid components separate into liquid-ordered l_o phases consist of saturated lipids and cholesterol and liquid-disordered l_d phases, the unsaturated lipid region [1–3]. Recently, strong evidence has been found for the existence of an interaction (or coupling) between both layers of lipid membranes. A comprehensive review of *in-vivo* experiments probing the existence of the coupling can be found in Ref. 4. Collins and Keller proposed that domains in one layer could either induce or suppress domain formation in the other layer [5]. These also support the existence of coupling between layers. The recent probable range of the coupling strengths is 0.1 - 0.2 in the thermal energy unit k_BT [6]. Interactions between lipids in the same layer can be represented by the unlike-nearest-neighbor interaction ω_{AB}, and the typical values of ω_{AB} lie between 100-300 cal/mol [7].

Experimental measurements of the domain growth dynamics, averaged domain size (R) versus time t, in lipid membranes have been reported in several previous papers. Interestingly, the results are the subject of lively debate. In 2006, Saeki *et al.* measured the domain dynamics in cell-sized liposomes and found the relationship $R(t) \sim t^{0.15}$ for off-critical mixtures [8]. The work by Yanagisawa *et al.* in 2007 found that two growth regimes could be observed in lipid domains [9]. Some domains grow in the form $R(t) \sim t^{2/3}$, which is called normal coarsening. On the other hand, some domains are suppressed after reaching some sizes. This mechanism is called the trapped regime. In 2010, Liang *et al.* measured domain growth dynamics in ternary lipid

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membranes having the ratio saturated lipid : unsaturated lipid : cholesterol equal to 1:1:1. They found that domains initially coarsen as $R(t) \sim t^{1/3}$; then, they grow faster as $R(t) \sim t$ [10].

Although lipid membrane structure has two lipid monolayers but there are only a few theoretical models that could capture this structure. A work by Laradji and Kumar seems to be the first work to address this point. They proposed a double-layer-type model, based on a dissipative particle dynamics (DPD), for a spherical membrane [11]. In 2006, they extended the model to the case of each layer having a different lipid composition; such membranes are named asymmetric membranes [12]. Later, the double-layer-type model received more attention. Frazier *et el.* [13] proposed lattice Monte Carlo (MC)-based models while the mean-field-based models were proposed by Putzel and Schick [14] and by Wagner *et al.* [15]. However, although asymmetric membranes can be created in laboratories, until the present, no experimental or theoretical works have explored the effect of lipid mismatch areas on the domain growth kinetics.

Recently, the Ising model has been proposed as a candidate to describe the phase separation in a lipid membrane [16–18]. From the computational point of view, the Ising model is straightforward to implement and requires fewer computational resources than a molecular-dynamics-based model. The length and the time scales obtained from the Ising model are also larger than these obtained from the molecular-dynamics-based model. If the domain growth dynamics is to be investigated, the length and the time scales are important issues. The Ising bilayer model has been studied for a long time [19, 20]. However, its application to find the domain coarsening dynamics in lipid membranes has not been considered until the present.

In the present work, we propose an Ising bilayer with lipid interactions, obtained from recent experiments, for modeling a lipid membrane. Our goals are to investigate the effect of the mismatch area on the domain growth dynamics by means of Monte Carlo simulations. The rest of this paper is schematically organized as follows: In Sec. II, a model of the lipid bilayer is presented. In Sec. III, the details of the simulations are described. In Sec. IV, the results from the MC simulations are presented and discussed. In Sec. V, finally, our findings are summarized.

II. MODEL FORMULATION

In the present work, a lipid bilayer is modeled as two-coupled Ising models, as shown in Fig. 1. The outer and the inner lattices represent the outer and the inner layers of lipid membranes, respectively. Each lattice is composed of a binary mixture of two particles (A and B) that represent two liquid phases: l_o and l_d. Particle A is a condensed complex between a saturated lipid and cholesterol (l_o phase). The unsaturated lipid is represented by particle B (l_d phase). To represent a three-lipid system as an effective binary mixture, we use the experimental fact that cholesterols tend to pack with saturated lipids to form l_o domains separating from the unsaturated lipids (the matrix phase). The effective binary mixture assumption is also used in previous works [15,21].

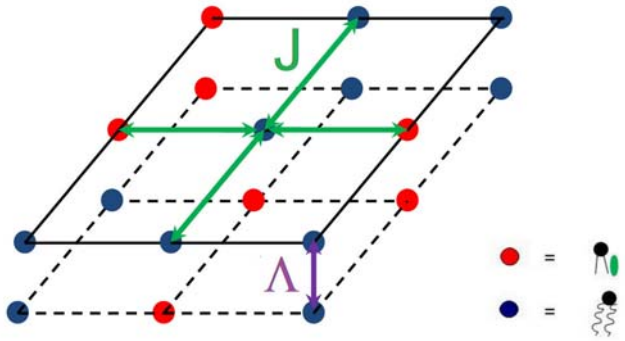

Fig. 1. (Color online) Schematic of the lipid bilayer used in this work. The outer and the inner lattices represent the outer and the inner layers of the lipid membranes, respectively. The intra-layer interaction between nearest-neighbor spins is denoted by J, and the inter-layer interaction of a pair of spins localized at the same positions on opposing monolayers is denoted by Λ. Packs of saturated phospholipid and cholesterol molecules are represented by red particles. Unsaturated phospholipid molecules are denoted by blue particles.

The Hamiltonian of this system is a combination of the Hamiltonians corresponding to the outer layer H^{outer}, the inner layer H^{inner}, and the inter-layer interaction H^{Λ}. The total Ising-like Hamiltonian of the system is, thus, given by

$$\begin{aligned} H^{tot} &= H^{outer} + H^{inner} + H^{\Lambda} \\ &= -J \sum_{<ij>} S_i S_j - J \sum_{<ij>} \sigma_i \sigma_j - \Lambda \sum_i S_i \sigma_i \quad (1) \end{aligned}$$

where the total Hamiltonian of the system is H^{tot}, and S_i and σ_i assign spins that reside on the outer and the inner monolayer at site i, respectively. The values of the spins represent the types of particles. Particles A and B are represented by spins $+1$ and -1, respectively. The summations in the first two terms denote the sum of nearest-neighbor spins in the same layer, where the intra-layer interaction strength is J. The effect of the inter-layer interaction is represented by the parameter Λ. The summation in the last term denotes the sum over all sites. The values of J and Λ are constrained to be positive, corresponding to the preferential affinity of the same components. The coupling term within the Hamiltonian promotes parallel spin directions (or identical particle types) in both layers. Equal values of spins at the same lattice sites in the two-coupled monolayer lattices cause the energy of the systems to decrease.

III. SIMULATION DETAILS

In our systems, we assume that no particles enter or leave from two opposed lattices and that particles are not allowed to transfer between layers. Thus, the numbers of particles A and B on both layer are conserved. Another assumption is that the lattices we are considering are planar lattices. The well-known conserved-order-parameter Kawasaki dynamics is used in all simulations [22]. For Kawasaki dynamics, the acceptance ratio is expressed in this form

$$A(\mu \to \upsilon) = \begin{cases} \exp[-\beta \Delta H^{tot}] & \text{if } \Delta H^{tot} > 0 \\ 1 & \text{otherwise} \end{cases} \quad (2)$$

where $\beta = 1/k_B T, k_B$ is the Boltzmann constant and T is the temperature. The energy change after exchanging a pair of neighboring particles can be expressed by

$$\begin{aligned} \Delta H^{tot} &= H_\nu^{tot} - H_\mu^{tot} = \Delta H^{outer} + \Delta H^{inner} + \Delta H^{\Lambda} \\ &= 2J[S_k^\mu \sum_{\substack{i \neq k \text{ and} \\ \text{n.n. to } k}} S_i^\mu + S_{k'}^\mu \sum_{\substack{j \neq k \text{ and} \\ \text{n.n. to } k'}} S_j^\mu] \\ &\quad + 2J[\sigma_k^\mu \sum_{\substack{i \neq k' \text{ and} \\ \text{n.n. to } k}} \sigma_i^\mu + \sigma_{k'}^\mu \sum_{\substack{j \neq k \text{ and} \\ \text{n.n. to } k'}} \sigma_j^\mu] + \Delta H^{\Lambda}, \end{aligned} \quad (3)$$

where k and k' are a pair of adjacent spins that are proposed to exchange positions [23]. The summations are carried out over nearest-neighbor spins, except for the positions of k and k'. ΔH^{Λ} is a difference in inter-layer coupling energies before and after exchanging a pair of spins. In order to perform the moving processes, an attempt to swap between neighboring particles is applied to each layer at a time. Therefore, when a pair of spins was swapped Eq. (3) was computed with one of its first two terms is zero. For example, if we consider the particle exchange in the outer layer, only the first and the third terms are required for calculation. The second term is zero because no pair of particles in the inner layer is exchanged at that time. These processes are also applied when we consider particle exchange in the inner layer. To do so, we have to define one exchanging cycle as two attempts to exchange particles on both layers.

The interactions of a particular particle can be divided into two types: intra- and inter-layer interactions (J, Λ). The intra-layer interaction,J, is the interaction between a particle and its neighboring particles on the same lattice. The inter-layer interaction, Λ, is the interaction between two-opposed particles located on two lattices, as shown in Fig. 1. In order to connect the simulations to real-world problems, we compiled the values of J and Λ from experiments. An excellent review of the lipid/lipid and the lipid/cholesterol interaction strengths can be found in Ref. 7. The strength of the intra-layer interaction in that paper is called "unlike nearest-neighbor interaction, ω_{AB}". This parameter represents the mutual interactions between two types of lipids A and B. The unlike nearest-neighbor interaction is expressed by

$$\omega_{AB} = g_{AB} - \frac{1}{2}(g_{AA} + g_{BB}), \quad (4)$$

where g_{ij} is the Gibbs free energy if there is an interaction between molecules i and j. A positive ω_{AB} means that molecules A and B have an unfavorable interaction, leading to phase separation. On the other hand, a negative (or zero) ω_{AB}means that lipids A and B will mix together. There are a number of methods, such as nearest-neighbor recognition (NNR) and isothermal titration calorimetry methods (ITC), to calculate ω_{AB} from experimental data [24,25]. The absolute values of ω_{AB}, which were calculated using regular solution theory, are typically between 100-300 cal/mol in absolute values; rarely do they exceed 300 cal/mol [7]. Some recent works use the unlike nearest-neighbor interaction to represent the interactions between lipid components [13, 26–29]. Almeida's article also reviews the value of the inter-layer interaction. It is approximated to be 100 cal/mol, in the absolute value, for many cases. This implies that the intra-layer interactions of bilayers are typically larger than the inter-layer interaction.

On the theoretical side, estimates of the inter-layer interaction strength are rather confusing in magnitude. Collins [30] calculated the interaction from the line tension and got values around 0.1 - 1 $k_B T$. Risselada and Marrink obtained values of 0.1 - 0.2 $k_B T$ from molecular dynamics simulations [31]. Note that, if we convert the value obtained from experiments (100 cal/mol) to $k_B T$ units at room temperature, we will get a value around 0.16. This value lies within the estimated range of Rissalada and Marrink. In our simulations, the value of J has been set to 0.5 $k_B T$,which is larger than the critical value obtained from the famous Onsager solution ($J_C \approx 0.441\ k_B T$) [32]. This value ensures that phase separation must occur in the simulations. The value of Λ has been set to 0.15 $k_B T$. There are two reasons behind this value. The first reason is that this value lies in the range of the experimental and the theoretical calculations. The second reason comes from the experimental fact that the strength of the inter-layer interaction ($\sim$100 cal/mol) should be less than that of the intra-layer interaction ($\omega_{AB} \in (100 - 300)$ cal/mol). In this work, we relate J in the simulations with ω_{AB} from experiments. Therefore, the value of J must be larger than the value of Λ. Note that, the value $J = 0.5\ k_B T$ lies in the range of ω_{AB} obtained from Table 1 in Ref. 7.

Three systems are considered in our work. System I represent a symmetric lipid system A:B = 50:50 on both monolayers. System II represent the system in which A:B = 30:70 (outer layer) and A:B = 50:50 (inner layer). System III represent the system in which A:B = 10:90 (outer layer) and A:B = 50:50 (inner layer). We can see that, theoretically, system I has no mismatch areas while system III has more mismatch areas than system II. Simulations have been performed on lattices of size $N_x \times N_y = 512 \times 512$ with a periodic boundary condi-

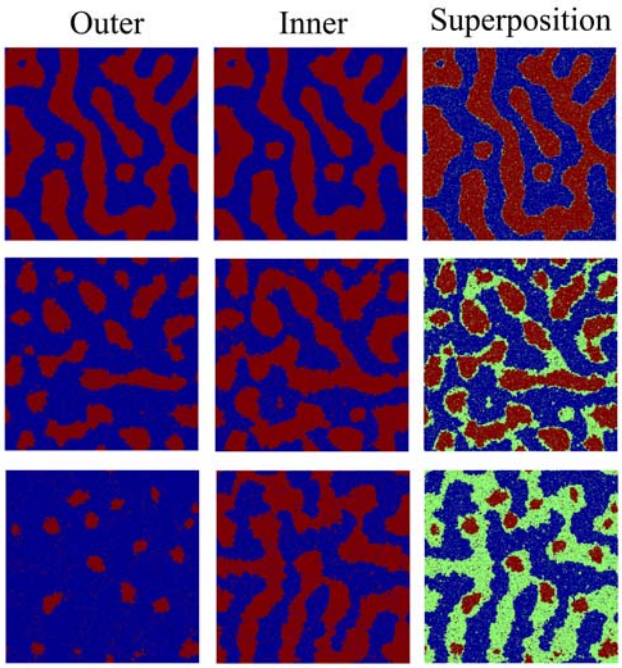

Fig. 2. (Color online) Snapshots of the phase morphologies of the symmetric (first row, system I) and the asymmetric (second row (system II), third row (system III)) structures at 3×10^6 MCS. The phase separations of the components in the outer and the inner layers are shown in the first and the second columns, respectively. Particles A are shown in red, and particle B are shown in blue. The top view (both layers observed simultaneously) of the bilayer are shown in the third column. For the top view, the matched regions are marked in red (A/A) and blue (B/B) whereas the mismatched regions are marked in green (A/B or B/A).

tion. Initially at MCS = 0 (or Monte Carlo Step = 0), the spins on both lattices are placed at a very high temperature. Then, the systems are quenched to a temperature that exhibits phase separation. Simulations were performed until the typical domain size reached approximately 1/10 of the lattice width to prevent the finite effect. The time was calculated as the number of Monte Carlo steps (MCS). We defined one MCS to be equal to $N_x \times N_y$ attempts to perform one exchanging cycle.

IV. RESULTS AND DISCUSSION

Phase separation on both layers of the three systems is shown in Fig. 2. The first and the second columns represent domain configurations on the outer and the inner layers, respectively. Particles A and B appear in red and blue, respectively. The third column shows top views of the bilayers where the red and the blue regions represent matched (or registered) regions and the green regions represent mismatch (or unregistered) regions. For system I, the top row, the interactions between layers induce domains on both layers to arrange in similar patterns that can be clearly observed by using a superposition view. Most of the domains appear in red and blue. This situation was reported earlier in experiments performed by Veatch and Keller [1]. From those experiments, if both layers are prepared with the same lipid composition, domains on both layers are completely overlapped. Therefore, our model could reproduce this kind of domain patterns in experiments. For asymmetric cases, middle and bottom rows, the coupling induces domains of the same kind to align in opposite directions, but they cannot form a complete overlap. In the superposition views, red domains are observed within the mismatch regions (green regions). This situation was also reported in experiments by Collins and Keller [5]. They prepared asymmetric bilayers and observed three-phase regions (dark, medium and bright regions). From their explanation, three-phase regions were due to lipid alignment on both layers. A region that has saturated (unsaturated) lipid molecules on both layers will appear as dark (bright). On the other hand, it appears as a medium intensity when lipid types on both layers are different. Therefore, our model could reproduce these kinds of domain patterns shown in experiments performed on asymmetric lipid bilayers as well. Note that, the three-phase regions cannot be constructed by using a model that assumes the membrane to be a single-layer object.

Evolution of the averaged domain sizes for the three systems are shown in Fig. 3. We consider the effect of the mismatch area on the kinetics of domain growths through an increasing of mismatch areas. The averaged domain sizes, R, on one layer at a particular time, are calculated by using the circularly averaged structure facture, $I\,(k)$, which is expressed by

$$I(k) \equiv \frac{1}{N}\left|\sum_{r_i} e^{i\,\overrightarrow{k}\cdot\overrightarrow{r_i}}\, s(\overrightarrow{r_i})\right|^2, \tag{5}$$

where $s\,(\overrightarrow{r_i})$ represent the spin at position $\overrightarrow{r_i}$, and N is the total number of spins. In order to calculate the typical length scale of the system, we have to investigate the wave number dominating the system. The dominant wave number can be found by using the following expression [37]:

$$k \equiv \frac{\int_0^\infty kI(k)dk}{\int_0^\infty I(k)dk}. \tag{6}$$

From this method, the typical domain size (domain diameter) is approximated by $R = \pi/k^*$ [37]. Note the paper by Amar [40] used $2\pi/k^*$ as a definition of the typical domain size, but the difference in the multiplication number does not affect the domain growth rate.

We have found that domains on both layers, for each system, evolve at similar rates. The domains coarsen in the power-law form $R\,(t) \sim t^\alpha$, where α represent the growth exponent. Increasing mismatch areas causes

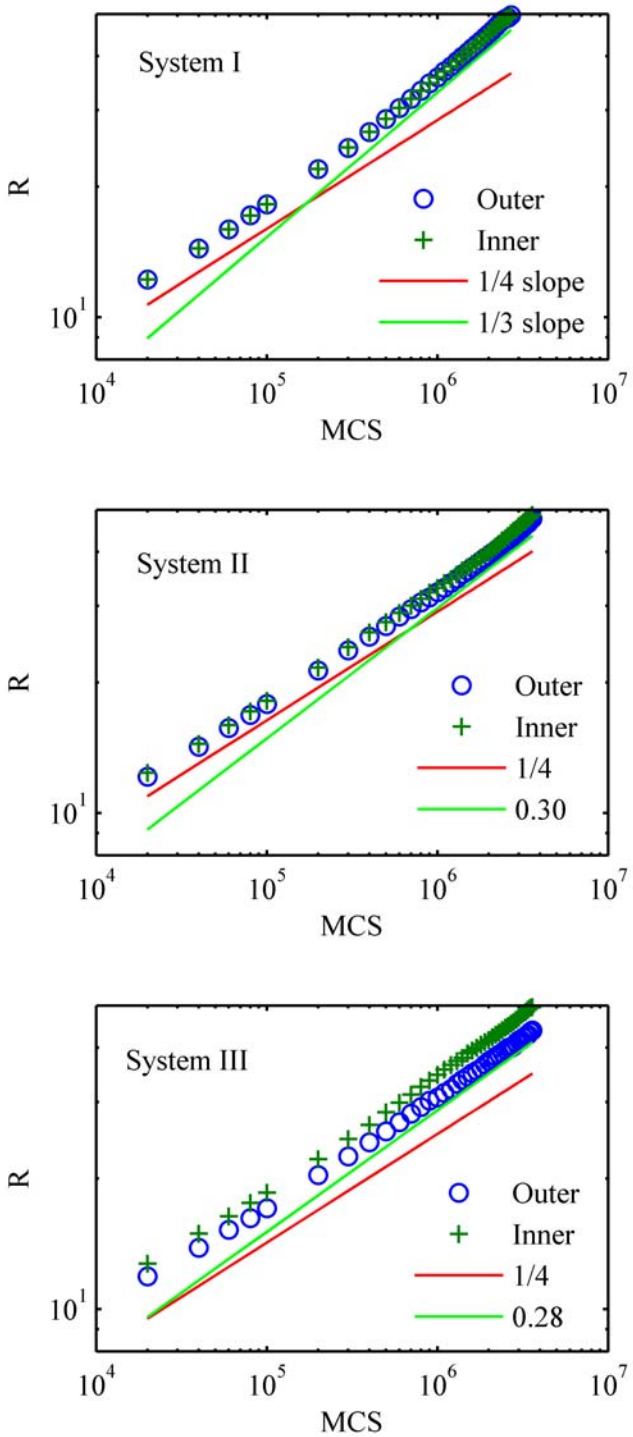

Fig. 3. (Color online) Comparison of the domain growth dynamics among the three systems. In all compositions, averaged domain sizes grow according to a power-law of time, $R(t) \sim t^{\alpha}$. For each system, domains of both layers grow at similar rates.

slower domain coarsening, which is shown clearly in the exponent values, α, shown in Table 1. The growth exponents in Table 1 were calculated by fitting data after $1{\times}10^6$ MCS, we assume that the domains are in a late state of growth, to the formula $R(t) = At^{\alpha}$. The growth exponent of system I is approximately 0.33 while the growth exponents of system II and III are around 0.30 and 0.28, respectively. The power-law forms of domain growth were also observed in earlier experimental and computational works [8–12,33–36], but the values of the growth exponents of those works are different. Our growth exponent values lie in the range of exponents obtained in the work of Ehrig *et al.* [35].

Table 1. Growth exponents for each layer of systems I-III.

system	α of outer layer	α of inner layer
I	0.3394 ± 0.0064	0.3393 ± 0.0063
II	0.3048 ± 0.0062	0.3094 ± 0.0073
III	0.2797 ± 0.0045	0.2812 ± 0.0049

The growth exponents reflect the mechanisms driving the diffusion of particles. Theoretically, the movement of particles can be divided into surface and bulk diffusion [38, 39]. Surface diffusion corresponds to the sliding of particles along the interface between two species without detaching from the domain. In this process, isolated domains move slowly, and domain form when two or more domains merge altogether. This process arises easily when individual domains are very close. Another type of diffusion is movements in which particles have a slight bias to leave smaller domains and join bigger domains. Small domains will get smaller and disappear eventually while the bigger domains will get bigger. This process can be divided into three events: detaching from their domain, diffusing in a region of the opposite specie and reattach to the domain of the same kind. However, one can consider only events of the diffusion of a particle in a region of opposite specie as bulk diffusion [33]. In this work, we follow this criterion. In the classical Ising model, at an early stage, there are many small domains in the system, and the domains are very close to each other. Thus, surface diffusion dominates the growth rates. At late time, the number of domains decreases, distances between domains are larger, and their sizes are relatively large compared with those at early time. Therefore, surface diffusion rarely occurs. This causes bulk diffusion to dominate the system at this time. One can estimate the domain growth rate for both types of diffusion by calculating the typical domain sizes at a particular time. The previous work by van Gemmert *et al.* found that the domains evolve with the rate $R(t) \sim t^{1/4}$ ($R(t) \sim t^{1/3}$) for surface diffusion (bulk diffusion) [39].

In order to investigate the effect of increasing mismatch area, due to the change in lipid ratios on the outer layer, on the domain kinetic on the inner layer, we count surface and bulk diffusion events occurring on the inner layer in each Monte Carlo cycle. To do so, we have to characterize the state of the particle (or spin) on site i by a number $m_i$of its neighbors that are aligned, which

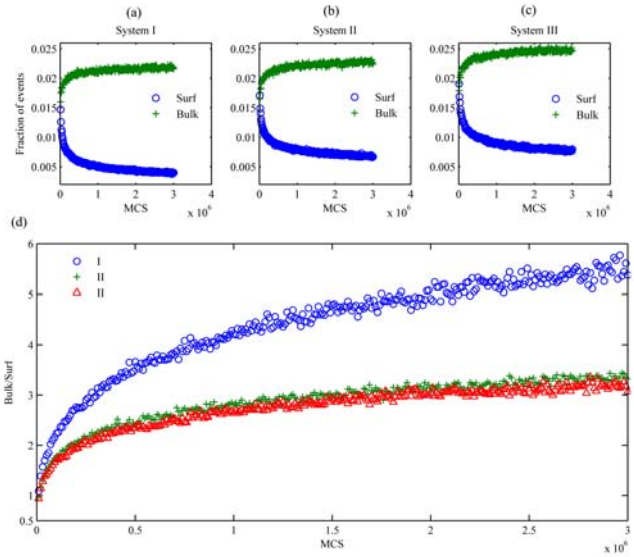

Fig. 4. (Color online) (a)-(c) Fraction of surface and bulk diffusion events for each Monte Carlo cycle occurring on the inner layer in systems I-III. (d) Ratios of bulk diffusion events to surface diffusion events corresponding to each Monte Carlo cycle occurring on the inner layer in systems I-III.

is defined by

$$m_i = \sum_{j\epsilon<ij>} \delta\left(s_i, s_j\right), \tag{7}$$

where s_i represent the spin on one layer at site i, the summation runs over neighboring sites of site i, and δ is the Dirac delta function. The number is usually called the coordination number of the particle (or spin) on site i [37,38]. On a square lattice, the exchange of two adjacent particles at sites i and j, corresponds to a transition from $\{m_i, m_j\}$ to $\{3-m_i, 3-m_j\}$. Precisely, surface diffusion occurs when the transitions $\{1,2\} \leftrightarrow \{2,1\}$ and $\{1,1\} \leftrightarrow \{2,2\}$ are made while bulk diffusion corresponds to the transition $\{0,3\} \leftrightarrow \{3,0\}$, when only free movement is considered. When a particle (spin) exchange is made, the reference particle, the particle for which we track the change in the coordination number, is chosen by the minority rule (choose a particle that has a smaller coordination number) [33]. The fraction of surface and bulk diffusion events in each Monte Carlo cycle is shown in Figs 4(a)-(c). In the figure, surface diffusion events decrease as time goes on because the interfacial lengths between two species decrease in time. On the other hand, bulk diffusion events increase in time because particles tend to leave small domains and diffuse to bigger domains. Because bulk diffusion events dominate systems at late time, the growth dynamics at late time should be close to 0.33. The ratio of bulk diffusion events to surface diffusion events on the inner layer in all three systems is shown in Fig. 4(d). In all systems, this fraction increases in time, which means bulk diffusion events dominate those systems at the late stage. However, the ratio in system I increases faster than that in the other two systems. This explains why the values of the growth exponents in system I are close to 0.33. It is interesting to note that, the ratio of bulk diffusion events to surface diffusion events in system II is slightly larger than that in system III but a detectable increase in the growth exponents is seen. However, the growth exponents of system II do not reach values around 0.33. This also suggests that the effect of surface diffusion events cannot be completely eliminated for ratios of bulk diffusion to surface diffusion events around 3 - 3.2 but it can be completely eliminated for ratios beyond 4.

To the best of our knowledge, our work seems to be the first attempt to investigate the effect of the mismatch area on the domain growth kinetics by using Monte Carlo simulations. However, the seminal works by Laradji and Kumar [11,12] considered a membrane as a double-layer object and measured the domain coarsening dynamics, but they did not study the effect of the increasing mismatch areas on the domain evolution. To the present, there is still no experimental measurement on the growth dynamics in asymmetric bilayers. Therefore, from our results, we can propose a possibility to detect the slowing down of domain growth dynamics when lipid mismatch areas increase. The most important role of lipid domain (or lipid raft) at the cell surface may be their function in signal transduction [41]. The domain provides a compartmentalization of cellular membranes to regulate transport and cell signaling processes. Therefore, understating lipid domain formation could greatly benefit our understanding in the signal transduction process.

V. CONCLUSION

In this article, we have applied the Ising bilayer model to study the effect of mismatch areas on the domain growth dynamics in lipid membranes. The probable values of the intra- and the inter-layer interactions were used in the model. The phase separation patterns of both layers were obtained by using Monte Carlo simulations. We have found that the model could reproduce domain patterns that are consistent with those observed in experiments and that the domain growth dynamics for the symmetric and the asymmetric cases follow a power-law of time. Domains on both layers of each system grow at similar rates. A slowing down of domain coarsening is observed when the mismatch areas are increased.

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